

# Convert Measurement Units

Lesson 4-6

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*1 foot = 12 inches, so the factor is 12*

### Conversion Factor

A ratio that helps you change one unit into another.

*Inches, feet, pounds, ounces*

### Unit

A standard amount used to measure, like an inch or a liter.

*2 feet = 24 inches*

### Equivalent

Having the same value, just in different units.

*12 inches per 1 foot*

### Ratio

A way to compare two amounts.

*12 inches = 1 foot, 3 feet = 1 yard, 16 ounces = 1 pound*

### Customary units

The U.S. way of measuring, like inches, feet, and pounds.

*1 meter = 100 centimeters, 1 kilogram = 1,000 grams*

### Metric units

A way of measuring by 10s, like meters, liters, and grams.

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can convert measurement units using ratios and conversion factors.**

#### 1 Notice

You're building a brand-new arcade cabinet!

#### 2 Key idea

I can convert measurement units using ratios and conversion factors.

#### 3 Words I'll use

Conversion Factor

Unit

Equivalent

Ratio

Customary units

Metric units



#### Think About It

- Why might the blueprint and the parts catalog use different units?
- How many inches are in 1 foot? How many feet in 1 yard?
- How do you decide whether to multiply or divide when converting?



#### Watch

Watch your teacher model one example. Jot what you see.

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#### We try

Solve the next one together as a class.

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#### You try

Now try one on your own.

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**The cabinet blueprint uses feet but the parts catalog uses inches. Why do you have to convert, and which is the bigger unit, a foot or an inch?**

#### Level 1 support

**Start here:** A foot is bigger than an inch, so I must convert to compare or combine the measurements.

#### Need a hint?

**Sentence starters (optional)**

**I have to convert because the blueprint uses \_\_\_\_ but the catalog uses \_\_\_\_.**

*Tengo que convertir porque el plano usa \_\_\_\_ pero el catálogo usa \_\_\_\_.*

**One \_\_\_\_ is bigger than one \_\_\_\_.**

*Un/Una \_\_\_\_ es más grande que un/una \_\_\_\_.*

**Word bank:** unit conversion factor customary units equivalent

**Listen for:** Students explain a foot is the larger customary unit (12 inches) and that mixed units must be converted to the same unit before working with them.

#### Level 2

**Predict: when you change feet into inches, will the number get bigger or smaller?  
Explain using the size of each unit.**

- The number gets \_\_\_\_ because \_\_\_\_.
- Switching to a smaller unit means \_\_\_\_.

## EXPLORE

**How did you use a conversion factor like '1 ft = 12 in' to fill in the ratio table, and how did you decide whether to multiply or divide?**

### Level 1 support

**Start here:** I multiply when going to a smaller unit and divide when going to a bigger unit.



**Need a hint?**

**Sentence starters (optional)**

**To convert** \_\_\_\_ **to** \_\_\_\_, **I** \_\_\_\_ **by** \_\_\_\_.

*Para convertir* \_\_\_\_ *a* \_\_\_\_, \_\_\_\_ *por* \_\_\_\_.

\_\_\_\_ **equals** \_\_\_\_ **because the conversion factor is** \_\_\_\_.

\_\_\_\_ *es igual a* \_\_\_\_ *porque el factor de conversión es* \_\_\_\_.

**Word bank:** (conversion factor) (unit) (equivalent) (ratio) (customary units)

**Listen for:** Students use the conversion factor as a ratio, multiply going to smaller units ( $5 \text{ ft} \times 12 = 60 \text{ in}$ ) and divide going to larger units ( $48 \text{ in} / 12 = 4 \text{ ft}$ ).

### Level 2

**Feet-to-inches uses multiply but inches-to-feet uses divide, even though both use '12.'**

**Explain why the same conversion factor leads to opposite operations.**

- I multiply when \_\_\_\_ but divide when \_\_\_\_.
- The same factor works both ways because \_\_\_\_.

CONNECT

A doorway is 80 inches tall and a new cabinet is listed as 7 feet. Your friend says it won't fit because 7 feet is 82 inches. Will it fit, and where is the error?

Level 1 support

**Start here:** 7 feet is 84 inches, which is taller than the 80-inch doorway, so it will not fit.

 **Need a hint?**

**Sentence starters (optional)**

7 feet = 7 x \_\_\_\_ = \_\_\_\_ inches.

7 pies = 7 x \_\_\_\_ = \_\_\_\_ pulgadas.

Since \_\_\_\_ inches is \_\_\_\_ than 80 inches, the cabinet \_\_\_\_ fit.

Como \_\_\_\_ pulgadas es \_\_\_\_ que 80 pulgadas, el gabinete \_\_\_\_ caber.

**Word bank:** conversion factor unit equivalent customary units

**Listen for:** Students convert 7 ft to 84 in (not 82), compare  $84 > 80$ , and conclude the cabinet will not fit; they catch the friend's arithmetic error.

Level 2

**How many inches would you need to trim off the cabinet to make it fit, and is that even possible for a real cabinet? Justify your reasoning.**

- You would need to remove \_\_\_\_ inches because \_\_\_\_.
- That is or is not realistic because \_\_\_\_.

## Guided Examples

### Example 1

**How many inches are in 7 feet?**

**Solution:**  $7 \times 12 = 84$  inches.

**Answer:** A. 84 inches

### Example 2

**How many feet are in 3 yards?**

**Solution:** 1 yard = 3 feet.  $3 \times 3 = 9$  feet.

**Answer:** A. 9 feet

### Example 3

**How many ounces are in 4 pounds?**

**Solution:** 1 pound = 16 ounces.  $4 \times 16 = 64$  ounces.

**Answer:** A. 64 ounces

## Write About the Math

### The Writing Revolution

I can explain my conversion using the words conversion factor, unit, equivalent, and ratio.

#### 1. Kernel Sentence subject + verb

**Model:** Unit is a standard amount used to measure, like an inch or a liter.

*Unidad es una cantidad estándar para medir, como una pulgada o un litro.*

**Write a kernel sentence about unit. Use a subject and a verb.**

*Escribe una oración base sobre unidad. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because · but · so

**Kernel:** Unit matters in math

*Unidad importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Unit matters in math because \_\_\_\_.**

*Unidad importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Unit matters in math, but \_\_\_\_.**

*Unidad importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Unit matters in math, so \_\_\_\_.**

*Unidad importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

#### Statement *Afirmación*

Tell one true fact about unit.  
*Di un hecho verdadero sobre unit.*

**Unit** \_\_\_\_.

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#### Question *Pregunta*

Ask a question about unit.  
*Haz una pregunta sobre unit.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

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#### Exclamation *Exclamación*

Show excitement about unit.  
*Muestra entusiasmo sobre unit.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

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#### Command *Mandato*

Tell a partner what to do with unit.  
*Dile a un compañero qué hacer con unit.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

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### 4. Explain Your Reasoning use a sentence starter

**I have to convert because the blueprint uses \_\_\_\_ but the catalog uses \_\_\_\_.**

*Tengo que convertir porque el plano usa \_\_\_\_ pero el catálogo usa \_\_\_\_.*

**One \_\_\_\_ is bigger than one \_\_\_\_.**

*Un/Una \_\_\_\_ es más grande que un/una \_\_\_\_.*

**To convert \_\_\_\_ to \_\_\_\_, I \_\_\_\_ by \_\_\_\_.**

*Para convertir \_\_\_\_ a \_\_\_\_, \_\_\_\_ por \_\_\_\_.*

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## Try It

Solve on your own. Check the answer key when you are done.

### 1. Convert 24 inches to feet.

- A. 2 feet
- B. 3 feet
- C. 12 feet
- D. 24 feet

Show your work:

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### 2. A shelf is 30 inches long. How many feet and inches is that?

- A. 2 feet 6 inches
- B. 3 feet 0 inches
- C. 2 feet 8 inches
- D. 1 foot 18 inches

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**A recipe says you need 5 feet of ribbon, but the store only sells ribbon in inches. Another recipe says you need 2 yards of fabric, but your ruler measures in feet. Convert both measurements and explain the steps you used. Which conversion factor did you use each time?**

*Sentence starter: 5 feet = \_\_\_\_ inches because \_\_\_\_ . 2 yards = \_\_\_\_ feet because \_\_\_\_ . The conversion factors I used were \_\_\_\_ .*

Show your work:

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## Reflect — Exit Ticket

**Which is the correct conversion of 5 yards to inches?**

- A. 180 inches
- B. 60 inches
- C. 15 inches
- D. 540 inches

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:** A. 2 feet —  $24 \div 12 = 2$  feet.
2. **Try It 2:** A. 2 feet 6 inches —  $30 \div 12 = 2$  remainder 6. So 30 inches = 2 feet 6 inches.
3. **Exit Ticket:** A. 180 inches —  $5 \text{ yards} \times 3 = 15 \text{ feet}$ .  $15 \text{ feet} \times 12 = 180 \text{ inches}$ . Or:  $5 \times 36 = 180$  inches.

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Unit is a standard amount used to measure, like an inch or a liter.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).